

Important Information:

Hostname: [REDACTED]

User: [REDACTED]

Command: \$ ssh [REDACTED]

Password: [REDACTED]

Wifi: [REDACTED]

Retrieving Camera Footage:

1. Connect to the appropriate network
2. Via terminal run \$ ssh [REDACTED]
**If you run into a warning: "WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!"*
 Run ssh-keygen -R [REDACTED]
3. Locate the appropriate .h264 file by picking the 8th to last video from when you first boot up the pi
4. Open an additional terminal, and run \$ scp [REDACTED]path/to/file ~/Downloads

```
rocketrycamera@wurocketry:~ $ ls
Bookshelf    Downloads    flightVid_4.h264  Public       Videos
cameraStuff  flightVid_1.h264  flightVid_5.h264  Templates
Desktop      flightVid_2.h264  Music            text.txt
Documents    flightVid_3.h264  Pictures         vidCount.txt
```

Figure 1: Your directory should look something like this

5. Run \$ brew install ffmpeg (once)
 Change directory (cd) to the file location, then run

```
$ffmpeg -framerate 30 -i flightVid x.h264 -c:v copy output.mp4
```

This step locally converts the file from .h264 into a .mp4 format for viewing.

Setup

Cron Tabs

Each Pi have crontabs running to make sure executeVideo runs at boot

1. \$ crontab -l verifies that the command @reboot sh cameraStuff/executeVideo.sh is running on the pi
2. \$ crontab -e is used to edit the file

Execute Video script

1. cd into "cameraStuff", and you can observe "executeVideo.sh" via nano or cat

```
#!/bin/bash

echo "Starting video recording"
FILE="vidCount.txt"
while true; do
    COUNT=$(cat "$FILE")
    COUNT=$((COUNT + 1))
    echo "$COUNT" > "$FILE"
    rpicas-vid -t 900s -o flightVid_$COUNT.h264
done
```

Figure 2: Contents of executeVideo.sh

This is what the file should look like exactly. It is simply a script that runs a while loop that creates a file, adds +1 to a text file called *vidCount.txt* to see how many video files have been created, and is also used to set the counter back to 0 before a launch.

Network

The Raspberry Pi is configured to connect to [REDACTED], however you can connect it to a mobile phone Hostpot as well. Currently it is only configured to [REDACTED] hotspot, however the steps below outline how to connect a new one.

1. Run `$ sudo nmtui`
2. Run `$ sudo nmcli device wifi rescan`
3. Utilize the GUI to 'Activate a connection', select your hotspot, and insert the correct name and password.
4. Verify your work with `$ nmcli connection show`

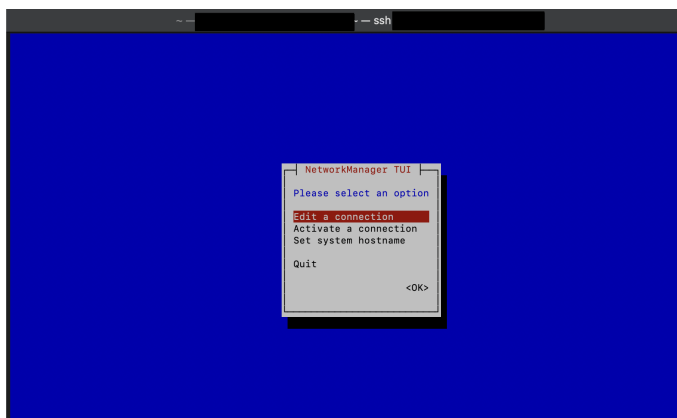


Figure 3: Network GUI

You should now be able to activate your hotspot when out of range of the [REDACTED] network, i.e. for a launch, it is convenient not to have to go back to WashU to retrieve the camera footage.